

Class 12: Genome Informatics Lab

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Q13. The sample size for each genotype is 108 for A/A genotype, 233 for A/G genotype, and 121 for G/G genotype. The median expression level of genotype A/A is 31.25, A/G is 25.06, G/G is 20.07.

```
c12 <- read.table("class12data.txt", header = TRUE, row.names=1)
# Sample size per genotype
summary(c12)
```

sample	geno	exp
Length:462	Length:462	Min. : 6.675
Class :character	Class :character	1st Qu.:20.004
Mode :character	Mode :character	Median :25.116
		Mean :25.640
		3rd Qu.:30.779
		Max. :51.518

```
table(c12$geno) #gives number of instances (sample size) of each genotype in geno
```

A/A	A/G	G/G
108	233	121

```
tapply(c12$exp, c12$geno, median) #creates median expression of each genotype
```

A/A	A/G	G/G
31.24847	25.06486	20.07363

Q14. The boxplot shows that A/A has a higher median ORMDL3 expression than G/G and A/G, with G/G being the lowest in median ORMDL3 expression. This means that the SNP does affect ORMDL3 expression as individuals with two G alleles have lower expression of ORMDL3 than individuals with 1 G allele and both are lower in expression than individuals with 2 A alleles and no G alleles. This indicates that the G allele is associated with decreased expression.

```
boxplot(exp~geno, data = c12, # makes a boxplot
        xlab = "Genotype", #makes the x label
        ylab = "ORMDL3 Expression Level", #makes the y label
        main = "ORMDL3 Expression by Genotype",
        col = c("lightblue", "lightgreen", "lightpink"))
```

